

1. Theoretical Analysis

Algebraic Decomposition of the Effective Channel. To build a fast algorithm, it is necessary to move from matrix multiplication operations to vector ones. Note that the matrix V_{eq} of size $L \times L$ can be represented as a weighted sum of outer products (rank-1 matrices) of the active rows:

$$V_{eq} = z \sum_{n=1}^N x_n (v_n^* v_n) = z \sum_{n=1}^N x_n M_n$$

Where v_n is the n -th row of the matrix V , and $M_n \in C^{L \times L}$ is the local contribution matrix of the n -th antenna.

Analysis of the Role of Coefficient z . The value of z is determined by only one antenna — the one whose own power (row norm) is the maximum among the active ones. Because of this, a straightforward greedy algorithm ("building the active antenna set from scratch") is insufficient: in the first steps, z will fluctuate randomly, disrupting the evaluation of the antennas' contributions.

Drawbacks of the Baseline Heuristics (H1 and H2)

- H1: Considers only the scalar power of the antenna, completely ignoring the signal phase and crosstalk (interference) between streams.
- H2: Ignores the antenna's own power and does not account for the influence of the coefficient z , which leads to significant signal degradation at high noise levels ($\sigma = 1$).

2. Algorithmic Solution: Backward Greedy

Based on the theoretical analysis, it is proposed to use a backward greedy algorithm (Backward Greedy Search). We start from a state where all N antennas are active, and iteratively remove one antenna at a time that makes the least useful contribution (or causes the most harm) until $K = N/2$ or $K = 3N/4$ antennas remain.

Step-by-step description of the algorithm:

1. Precomputations:

- Calculate the powers of all antennas $p_n = \|v_n\|^2$.
- Precompute the contribution tensor $M_n = v_n^* v_n$ for $n \in [1, N]$.
- Initialize the current sum matrix $S = \sum_{n=1}^N M_n$.

2. Loop (K times):

- For each active antenna i , consider the candidate matrix $S^{(i)} = S - M_i$.
- For each candidate, calculate the new coefficient $z^{(i)}$ in $O(1)$ time. (To do this, it is sufficient to track the indices of the two most powerful active antennas. If the first one is removed, z is determined by the second; otherwise, z remains unchanged).
- Calculate the test effective channel $V_{eq}^{(i)} = z^{(i)} \cdot S^{(i)}$.
- Evaluate the objective function $U_G^{(i)} = \det(V_{eq}^{(i)}(V_{eq}^{(i)})^* + \sigma I)$.

3. Update:

- Find the index i^* whose removal preserves U_G to the maximum extent (or even increases it due to interference reduction).
- Permanently remove antenna i^* ($x_{i^*} = 0$). Update $S = S - M_{i^*}$.

3. Computational Complexity

A "naive" brute-force implementation would require multiplying matrices of size $L \times N$ and $N \times L$ at each step, yielding a complexity of $O(NL^2)$ for a single evaluation step. For the complete greedy algorithm, this would result in $O(K \cdot N^2 \cdot L^2)$, which is unacceptably slow for large arrays.

The proposed algebraic decomposition reduces the complexity of evaluating a single antenna to $O(L^3)$ (calculating the determinant of an $L \times L$ matrix). The final computational complexity of the algorithm is $\mathcal{O}(K \cdot N \cdot L^3)$.